ASYNCHRONOUS OPTICAL COMPUTING SYSTEM

Reinaldo Javier Aguiar Marcano

March 11, 2026

Title: ASYNCHRONOUS OPTICAL COMPUTING SYSTEM

Inventor: Reinaldo Javier Aguiar Marcano, a citizen of the United States, residing at 2302 Britton Ridge Drive, Katy, TX 77494.

Cross-Reference to Related Applications

This application claims the benefit of and priority to U.S. Patent Application No. 19/559,705, the contents of which are hereby incorporated by reference in their entirety.

Abstract

A computing system that reduces or eliminates the need for a global, high-frequency clock signal by using an internal, massively parallel optical data fabric for communication. The system comprises multiple processing components (e.g., CPU cores, accelerators) interconnected by optical buses. Instead of operating on a strict, system-wide clock cycle, components operate asynchronously, triggered by the arrival of data frames over the optical fabric. The ultra-high bandwidth and low latency of the optical connections ensure that data arrives faster than a typical clock cycle, making a data-flow or event-driven architecture more efficient and eliminating the power consumption and design complexity associated with distributing a high-frequency clock signal across a chip or system.

Background of the Invention

Synchronous design, where all components in a system operate in lockstep with a central, high-frequency clock signal, is the dominant paradigm in modern computing. However, as systems become more complex, distributing this clock signal to every component without "clock skew" (where the signal arrives at different parts at different times) becomes a major design challenge and a significant source of power consumption. Asynchronous or "clockless" designs have been proposed as an alternative, but have been difficult to implement due to the challenge of coordinating components without a shared clock.

Summary of the Invention

The present invention is a computing system that enables practical asynchronous or quasi-asynchronous operation by using a massively parallel optical data fabric as its internal communication backbone. The key insight is that if the communication between components is fast enough, the arrival of data itself can serve as the trigger for computation, reducing the need for a separate, high-frequency clock signal.

The system comprises multiple computational units (CUs) connected by a network of the ultra-wide optical buses. Rather than waiting for the next tick of a global clock, a CU remains in a low-power state until an optical data frame arrives at its input port. The arrival of the frame acts as an event trigger, causing the CU to power up, process the data, and transmit the result over its output optical port to the next CU in the chain. Because the optical fabric can transfer massive data words in a single cycle, the system can achieve extremely high throughput using a data-flow model, where computation follows the availability of data, rather than being driven by an external clock. This reduces power consumption and eliminates the complexities of high-frequency clock distribution.

Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Federal Government and its military and defense agencies a perpetual, royalty-free, non-exclusive license to practice and use the invention described herein for any and all governmental, defense, and non-commercial purposes. This license explicitly excludes any and all state and local governments.

Brief Description of the Drawings

FIG. 1 is a conceptual diagram contrasting a traditional synchronous, clock-driven system with the asynchronous, data-flow system of the present invention.

Detailed Description of the Invention

Referring to **FIG. 1**, a comparison of architectures is shown. A traditional synchronous system (100) consists of multiple components (101a, 101b, 101c) connected by a bus. A central, high-frequency clock generator (102) distributes a clock signal (103) to all components, forcing them to operate in lockstep, which consumes significant power. The asynchronous optical system of the present invention (110) also comprises multiple computational components (111a, 111b, 111c), but they are interconnected by a high-bandwidth optical data fabric (112). There is no high-frequency global clock. Instead, computation is triggered by the arrival of data itself. Component (111a) processes data and sends a result as an optical frame (113) to component (111b). The arrival of this frame triggers (111b) to perform its operation, and so on, creating a data-flow cascade.

Claims

What is claimed is:

1. A computing system, comprising:
    - a plurality of computational units; and
    - a massively parallel optical data fabric connecting the plurality of computational units;
    - wherein at least one computational unit is configured to remain in a first power state and to enter a second, higher power state to perform a computation in response to receiving a data frame via the optical data fabric.
2. The system of claim 1, wherein the system does not include a global, high-frequency clock signal distributed to all computational units.
3. The system of claim 1, wherein computation is scheduled according to a data-flow model.

4. The system of claim 1, wherein the arrival of a data frame at a computational unit acts as an event trigger for that unit to perform an operation.

5. The system of claim 1, wherein the optical data fabric comprises a plurality of optical buses, each with a bit-width greater than 128 bits.

6. The system of claim 1, wherein the optical data fabric is an implementation of the interface described in U.S. Patent Application No. 19/559,705.

Drawings

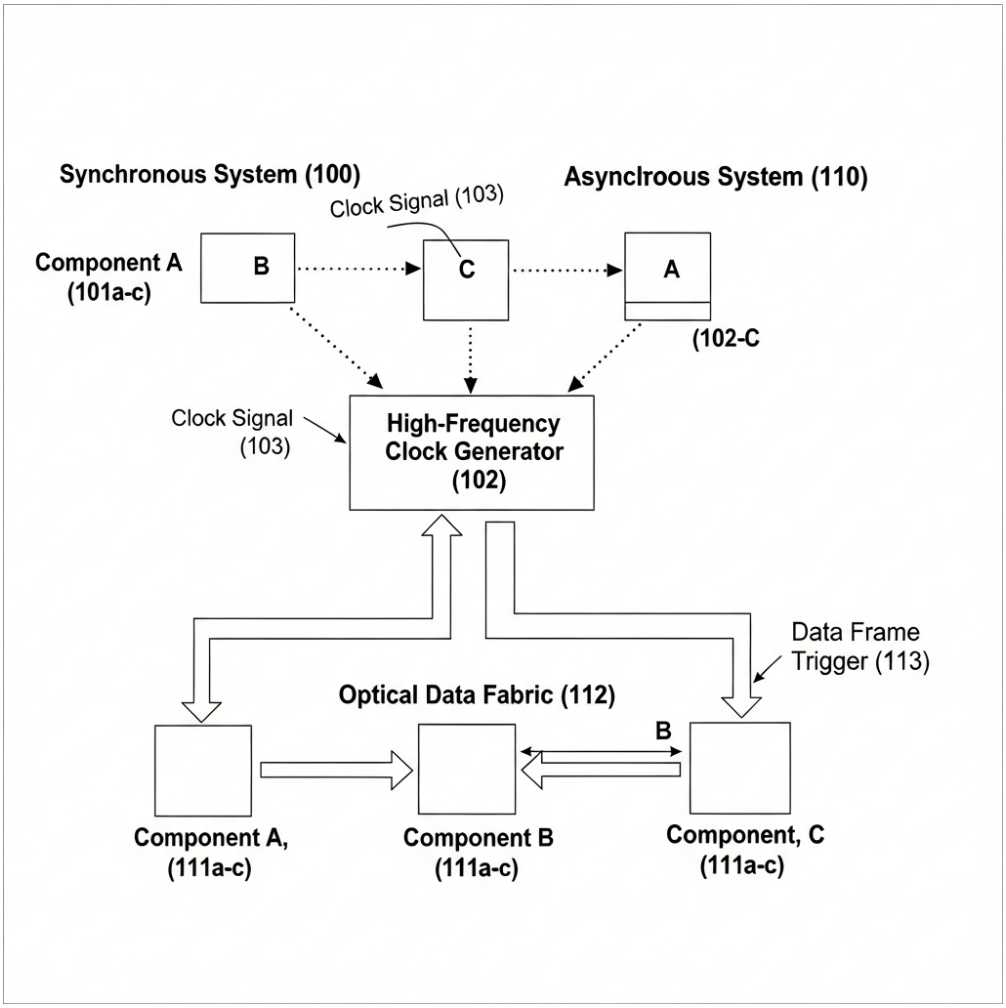


Figure 1: Conceptual diagram contrasting a traditional synchronous, clock-driven system with the asynchronous, data-flow system of the present invention.